On the Principles of Analytical Display

Inkwell

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Good information design relies on showing the data above all else. This handout demonstrates the Tufte layout through margin notes, sidenotes, margin figures, and full-width displays, all authored in Markdown and compiled through Inkwell.

Data-Ink and Visual Evidence

THE FUNDAMENTAL PRINCIPLE of analytical design is to show the data. Every bit of ink on a graphic requires a reason. If the ink does not tell the viewer something new, it should be erased.

Edward Tufte introduced the *data-ink ratio* in *The Visual Display of Quantitative Information* (1983).

Consider the ratio of data-ink to total ink in a graphic. A chart burdened with gridlines, redundant labels, and decorative hatching distracts from the numbers it purports to present. Maximize the data-ink ratio, within reason, and the result is a clearer picture.

The Anscombe Quartet

IN 1973, THE STATISTICIAN Francis Anscombe constructed four datasets that share nearly identical summary statistics yet look completely different when plotted.

Anscombe's quartet demonstrates why visualization matters: $\bar{x} = 9$, $\bar{y} \approx 7.5$, and $r^2 = 0.67$ for all four sets, yet the patterns differ radically.

The quartet drives home a simple lesson: always plot your data. Summary statistics alone can mislead.

"'{python display="code"} import os import matplotlib matplotlib.use("Agg") import matplotlib.pyplot as plt import numpy as np

x1 = [10, 8, 13, 9, 11, 14, 6, 4, 12, 7, 5] y1 = [8.04, 6.95, 7.58, 8.81, 8.33, 9.96, 7.24, 4.26, 10.84, 4.82, 5.68] x2 = [10, 8, 13, 9, 11, 14, 6, 4, 12, 7, 5] y2 = [9.14, 8.14, 8.74, 8.77, 9.26, 8.10, 6.13, 3.10, 9.13, 7.26, 4.74] x3 = [10, 8, 13, 9, 11, 14, 6, 4, 12, 7, 5] y3 = [7.46, 6.77, 12.74, 7.11, 7.81, 8.84, 6.08, 5.39, 8.15, 6.42, 5.73] x4 = [8, 8, 8, 8, 8, 8, 8, 19, 8, 8, 8] y4 = [6.58, 5.76, 7.71, 8.84, 8.47, 7.04, 5.25, 12.50, 5.56, 7.91, 6.89]

fig, axes = plt.subplots(2, 2, figsize=(6, 5), sharex=True, sharey=True) for ax, (x, y, label) in zip(axes.flat, [(x1, y1, "I"), (x2, y2, "II"), (x3, y3, "III"), (x4, y4, "IV")],): ax.scatter(x, y, s=20, color="steelblue", edgecolors="none") m, b = np.polyfit(x, y, 1) xs = np.linspace(3, 20, 50) ax.plot(xs, m * xs + b, color="gray", linewidth=0.8) ax.set_title(f"Set {label}", fontsize=10) ax.set_xlim(3, 20) ax.set_ylim(2, 14) ax.tick_params(labelsize=8)

fig.supxlabel("x", fontsize=10) fig.supylabel("y", fontsize=10) fig.tight_layout() os.makedirs(".inkwell/figures", exist_ok=True) plt.savefig(".inkwell/figures/anscombe.pdf", bbox_inches="tight") plt.close() "'

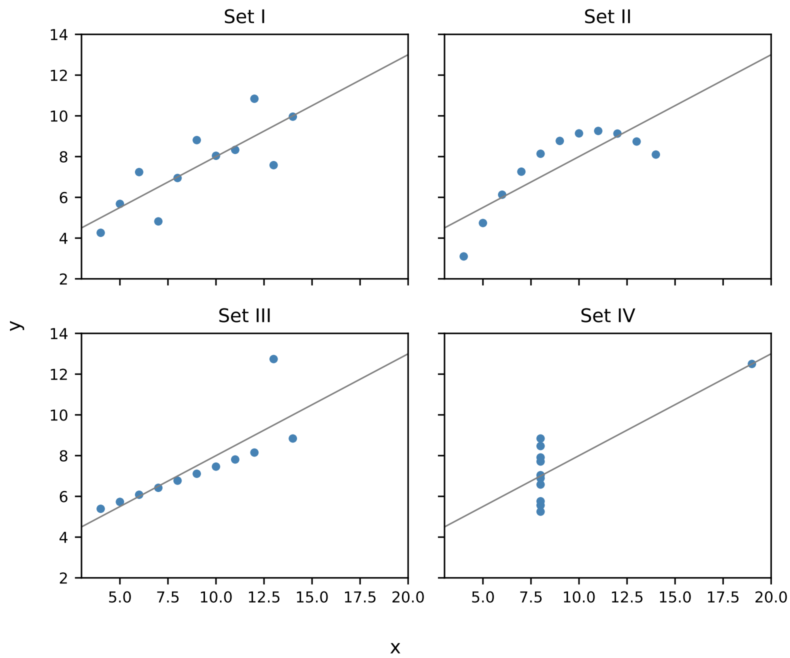

Figure 1: Anscombe's quartet: four datasets with identical summary statistics but distinct patterns.

As Figure 1 shows, the four sets share the same mean, variance, and correlation, yet the visual stories are entirely different.

Margin Figures and Full-Width Display

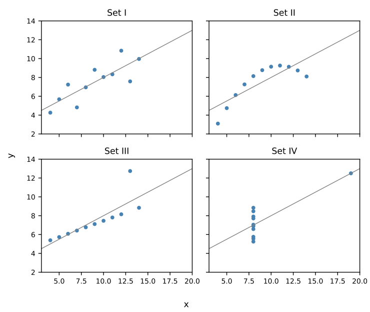

Figure 2: The same Anscombe quartet, placed in the margin for reference.

The wide margin of the Tufte layout serves as a parallel channel of communication. Figures, notes, and annotations in the margin let the main text flow without interruption.

Standard figures appear in the main column with their captions set in the margin by the Tufte class. Margin figures (above) fit small supporting graphics alongside the narrative.

Full-Width Sections

When a table or figure demands the full page width, the `fullwidth` environment extends into the margin area. This is useful for wide tables or panoramic figures that lose clarity when constrained.

Table 1: Summary statistics of the Anscombe quartet.

Dataset	n	$\bar{x}$	$\bar{y}$	s_x	s_y	r
I	11	9.0	7.50	3.32	2.03	0.816
II	11	9.0	7.50	3.32	2.03	0.816
III	11	9.0	7.50	3.32	2.03	0.816
IV	11	9.0	7.50	3.32	2.03	0.816

Mathematical Notation

CLEAR MATHEMATICAL typography supports analytical reasoning. The sample Pearson correlation coefficient is:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{1}$$

Equation 1 gives $r \approx 0.816$ for each of the four Anscombe sets, reinforcing the point that correlation is not causation, and that graphs are not optional.